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Santa Clara University	ELEN 164 Winter 2026	Shoba Krishnan
Laboratory #1: Passives: Introduction and current-voltage relationships		

I. Objectives

- To study the relationship between current and voltage in resistors, inductors and capacitors.
- To measure the component values of passives

III. Laboratory Procedure:

Part 1: Studying i-v relationship in resistive, inductive and capacitive circuits

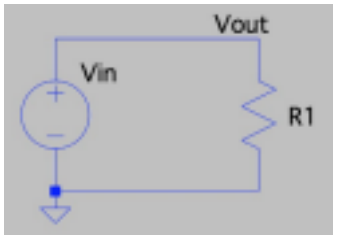


Figure 1(a): Resistive Load

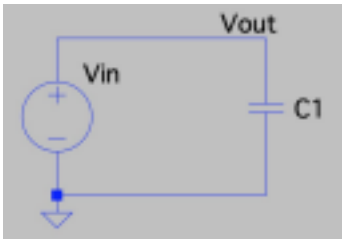


Figure 1(b): Capacitive Load

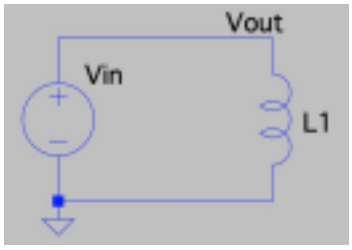


Figure 1(c). Inductive Load

Section 1:

Simulations:

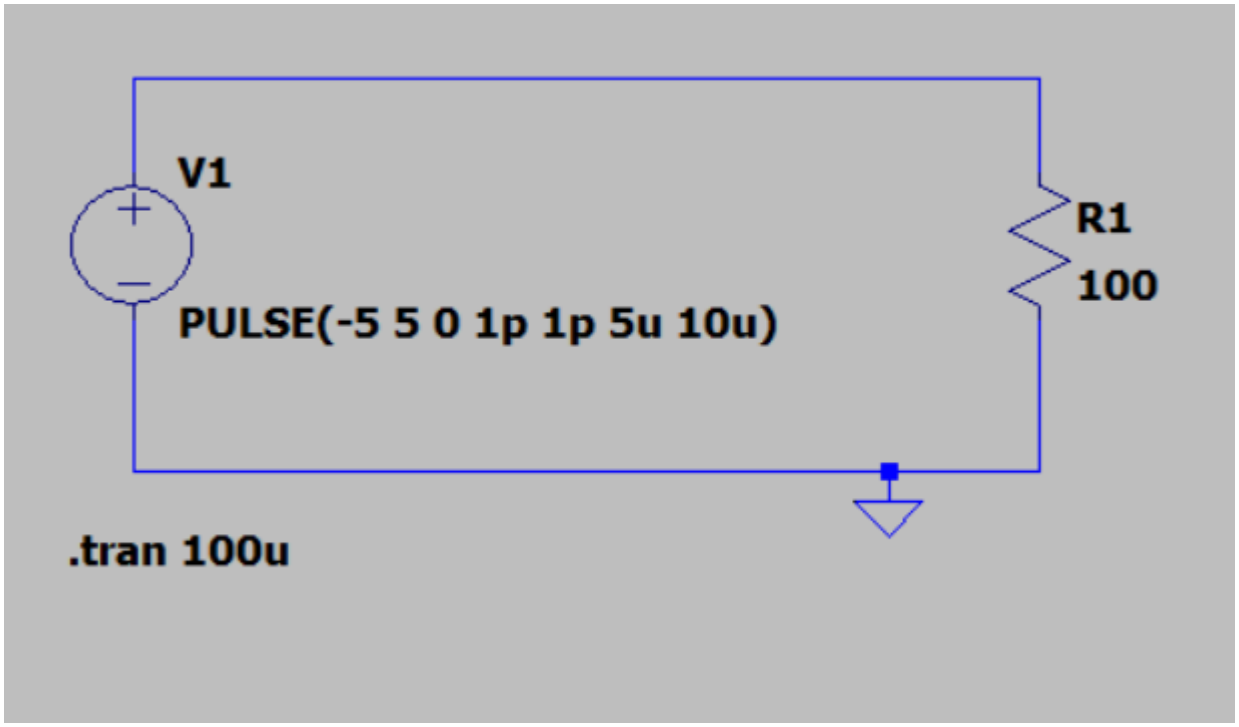
For each of the circuits in Figure 1 we will be studying the response of these passive elements to step/pulse inputs.

(1) Set the voltage source Vin to a square wave that switches between -5V and +5V at 100kHz with a 50% duty cycle. Use 1ps rise and fall times.

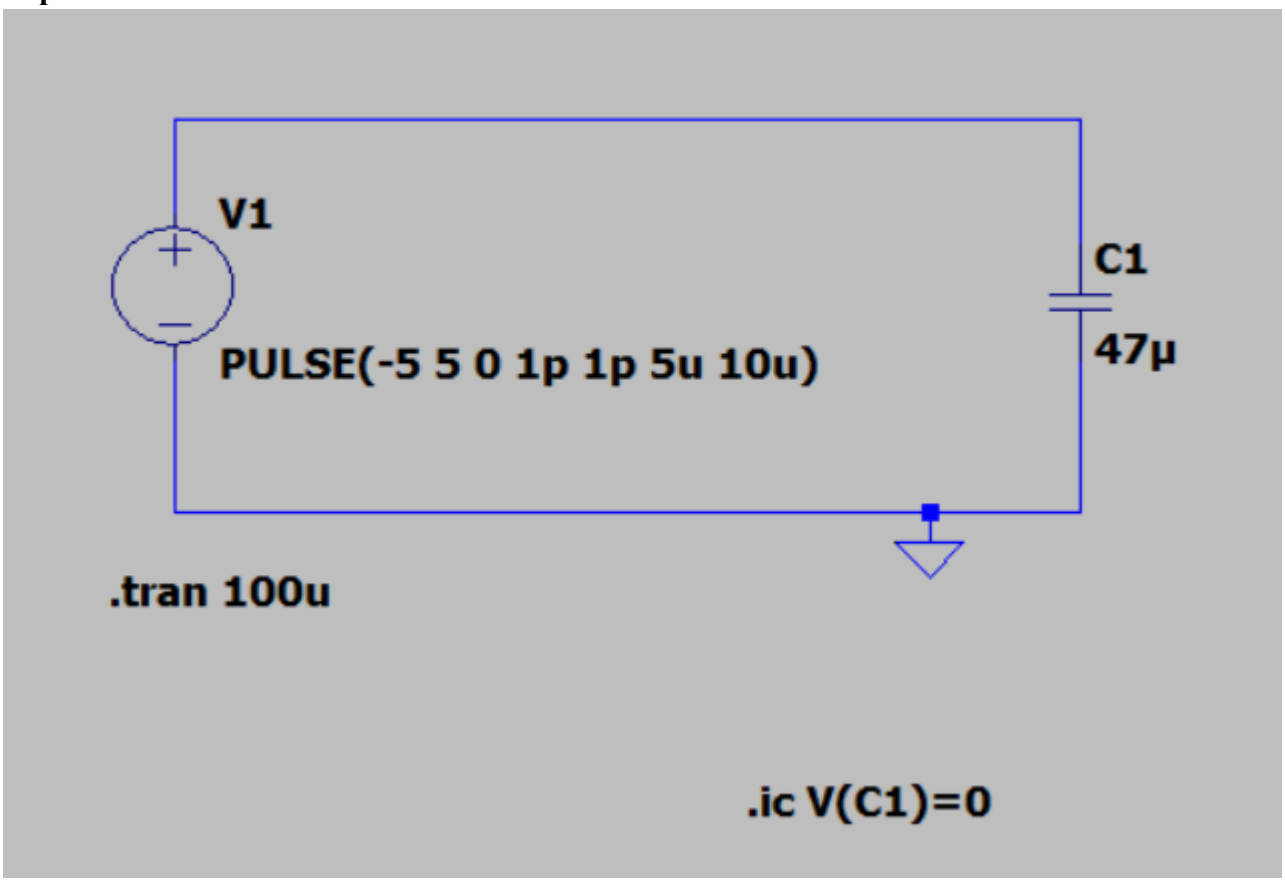
(2) Use values of R1 =100 ohms, C1 = 47 uF and L1= 50mH.

(3) Make sure to set the initial condition for the passive elements if needed. For example, setting inductor current to 0 is done by using the spice directive ".ic I(L1)=0".

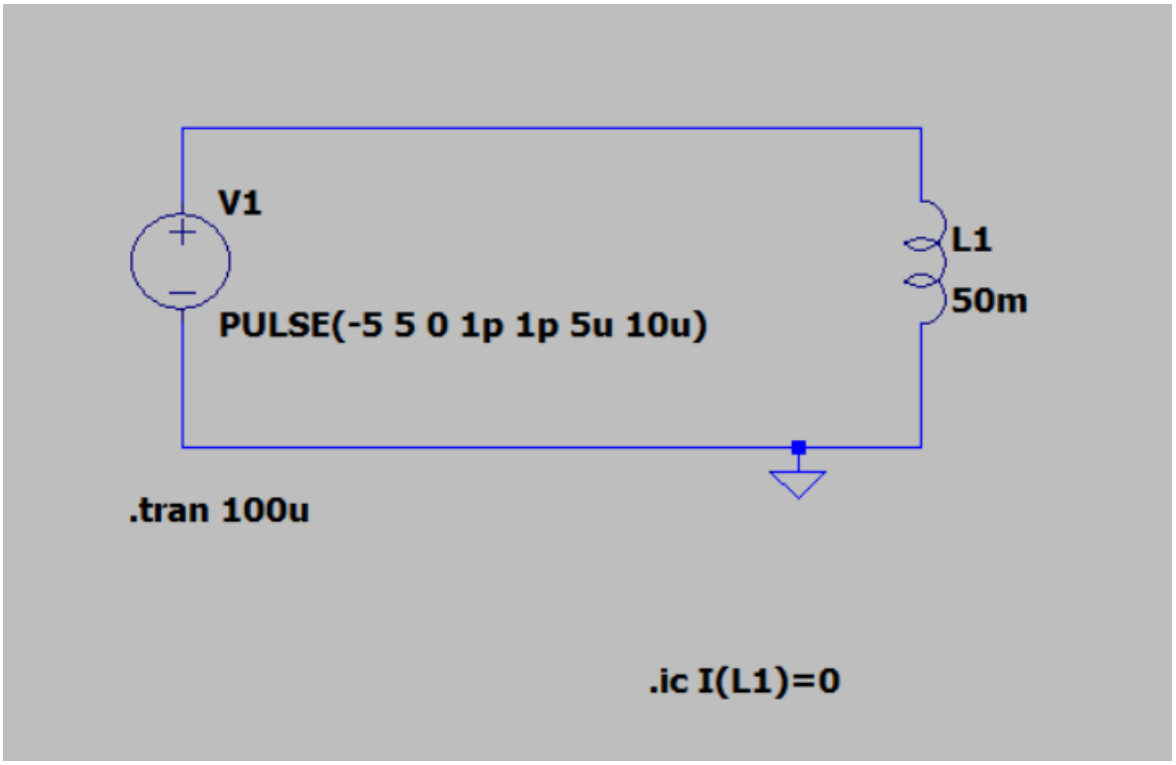
Resistor Circuit



Capacitor Circuit

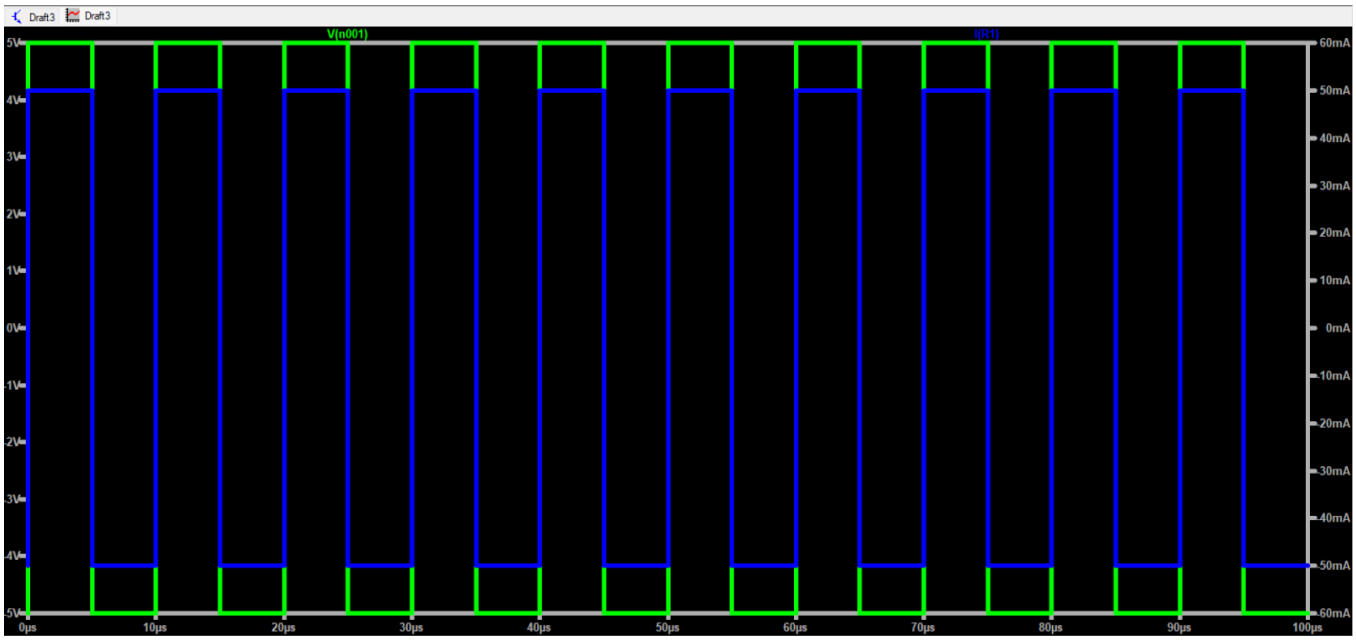


Inductor Circuit

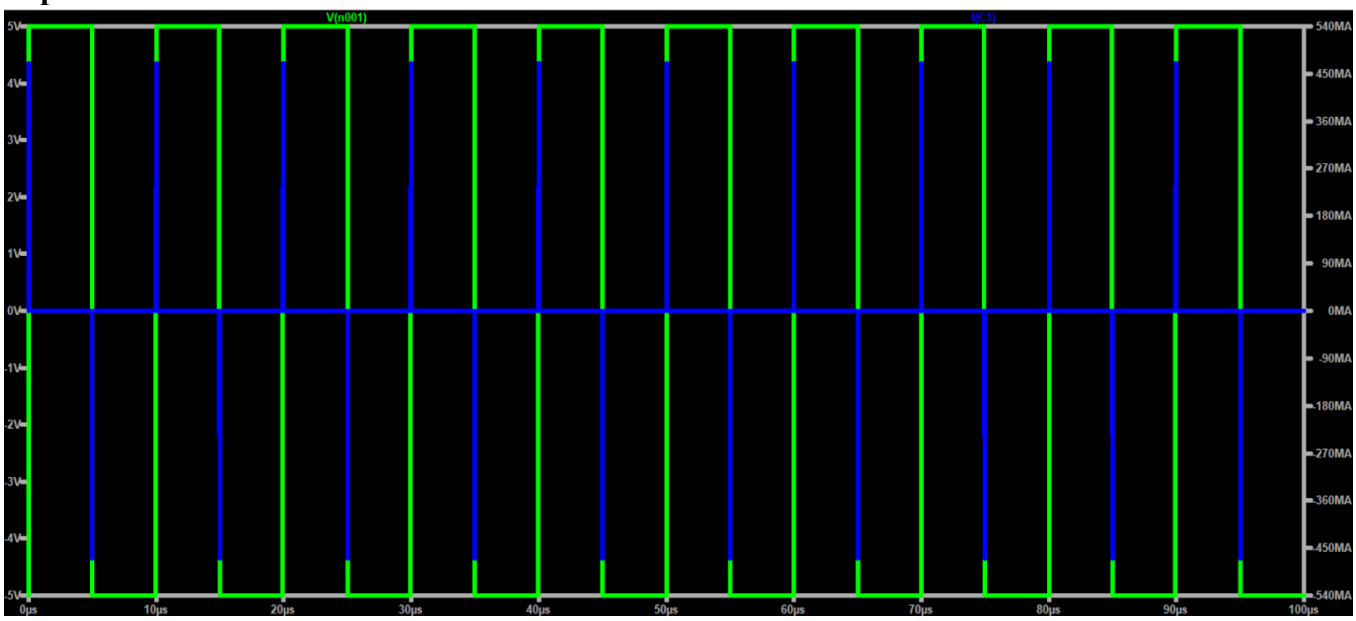


(4) Run a transient for 100us.
(5) Plot voltage and current in the passive element.

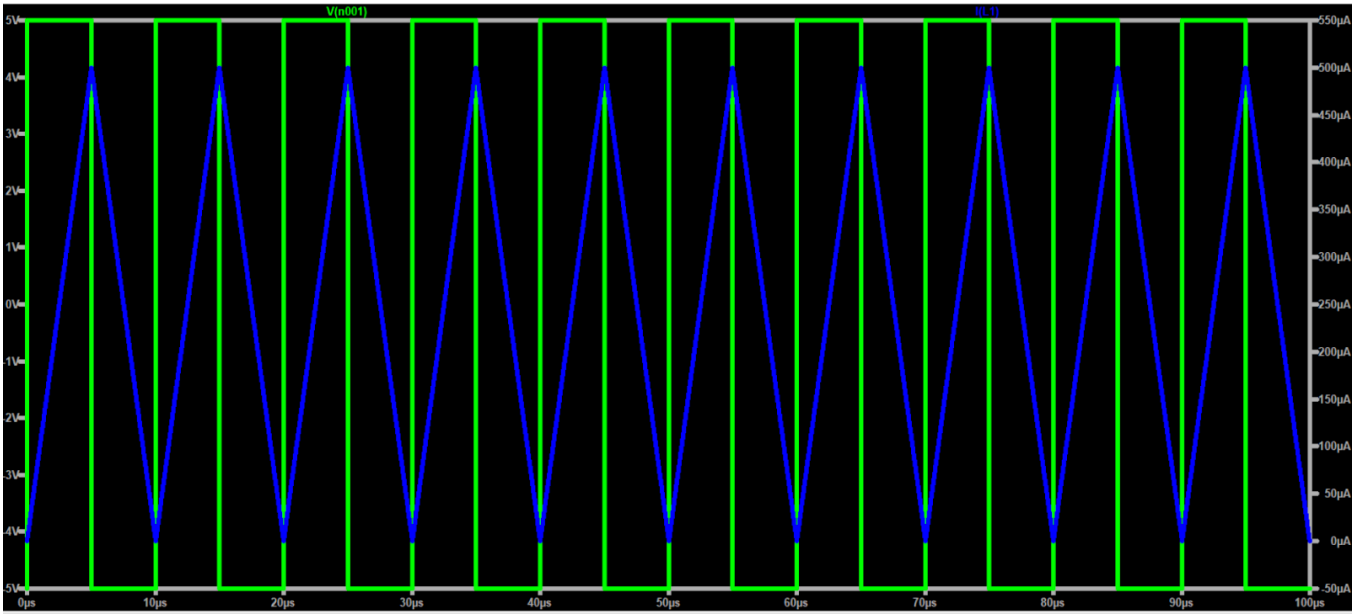
Resistor Circuit:



Capacitor Circuit:

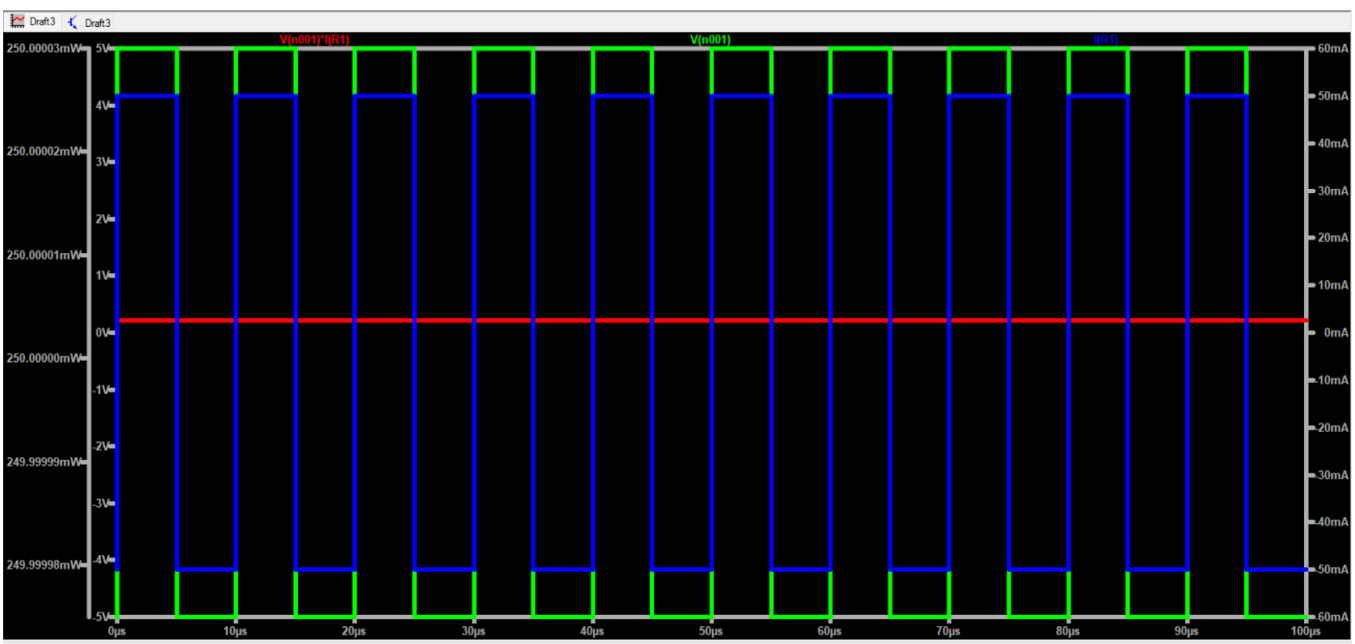


Inductor Circuit:

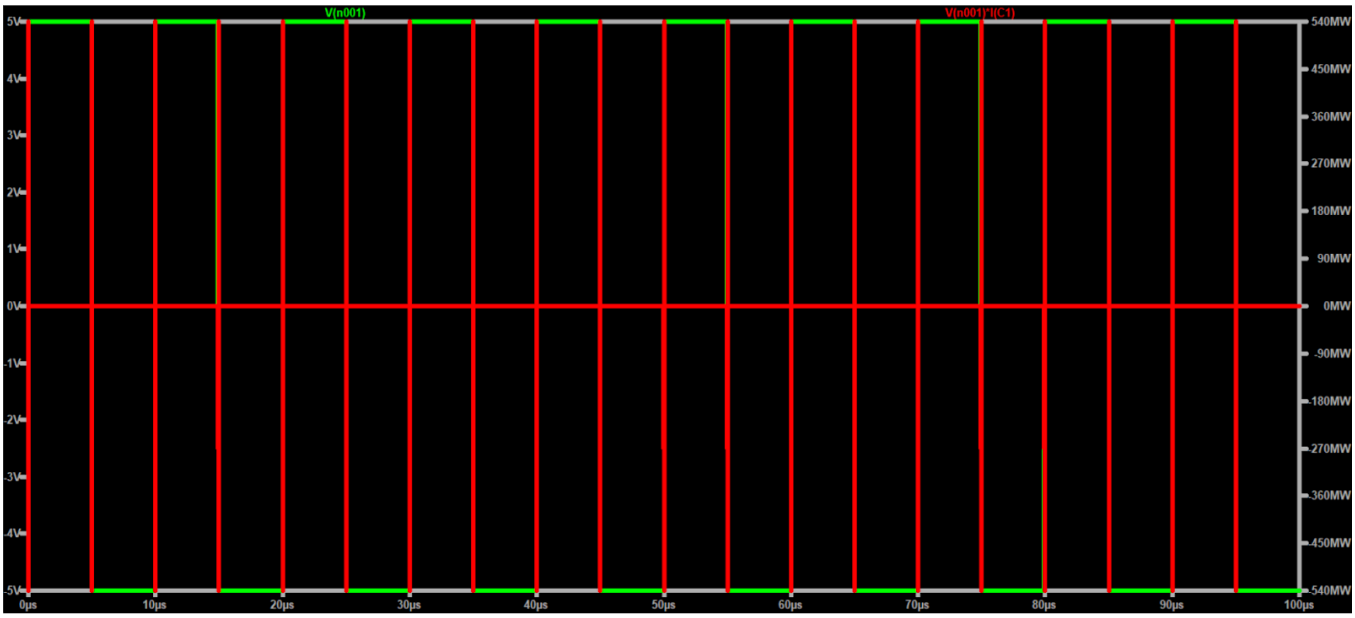


(6) Multiply the voltage and current and observe the instantaneous power.

Resistor Circuit:

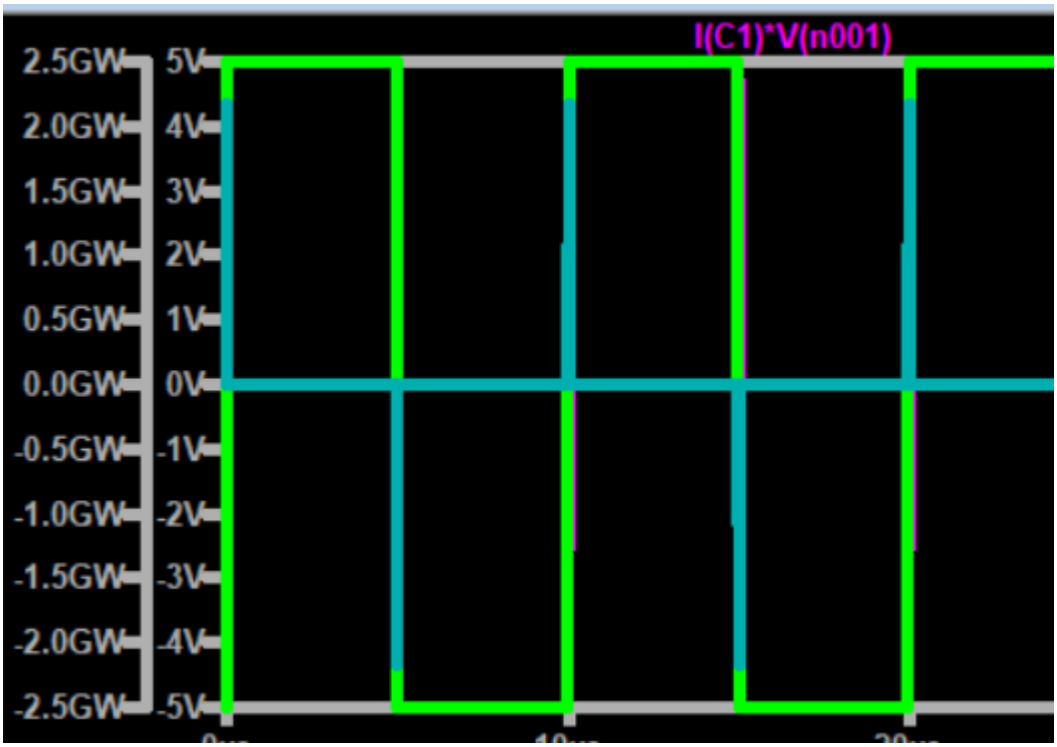


Capacitor Circuit:

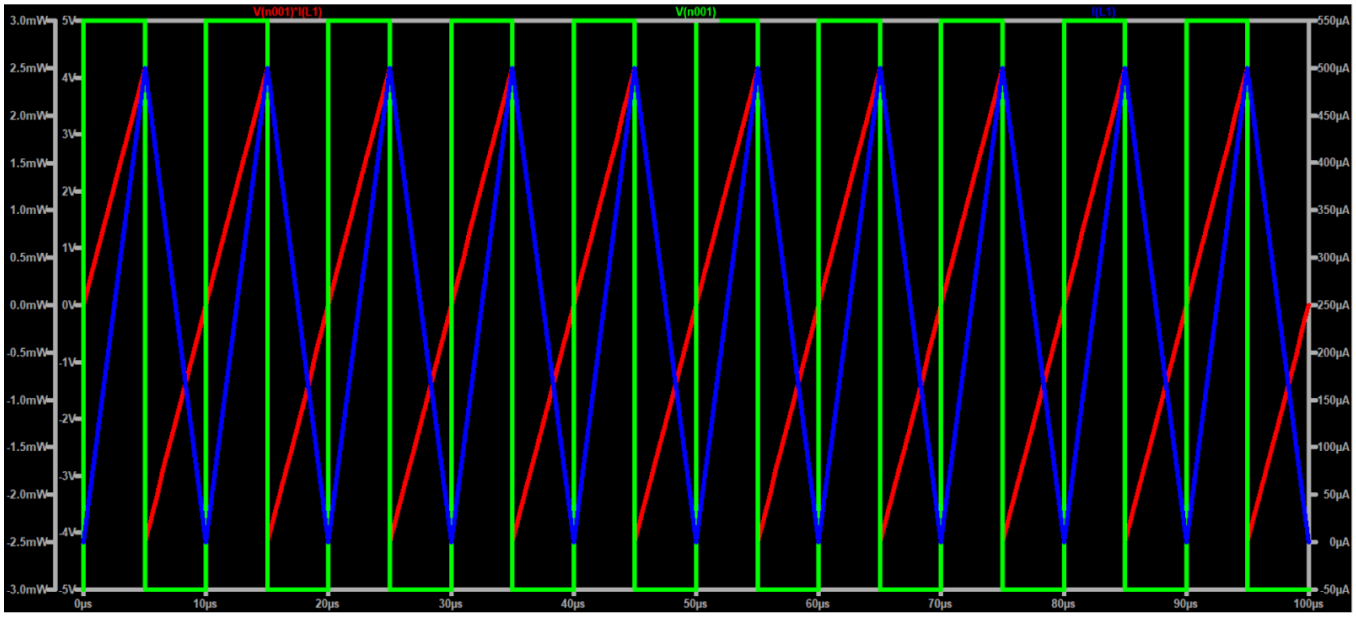


^^^current deleted to see power trace

Actual power numbers:



Inductor Circuit:



Observations:

(1) What is the shape of the current waveform?

Resistor: The same as voltage – the pulse wave.

Capacitor: Its a flat line at zero with jumps up or down at the same time as the voltage pulses.

Inductor: The current is a linear line with its slope being positive or negative depending on whether the voltage is positive or negative.

(2) Why does the current follow this pattern?

Resistor: V=IR

To satisfy the equation, the current must scale up to the voltage with a constant value.

Capacitor: I=Cdv/dt

To satisfy the equation, the current is dependent on dv/dt, which is 0 except for the spikes.

Inductor: V=Ldi/dt

To satisfy the equation, the change in current over time (slope) must equal V/L which (in this case) is a non-negative real number. Thus we see actual slope changes.

(3) Define the relationship for current vs. voltage for the element and support your explanations.

See above lol

(4) Check the peaks of the instantaneous power and explain their values.

Resistor: 250mW, which makes sense as V*I = Power and 5 * (50mA) = 250mW.

Capacitor: dv/dt = 10/(10^-12) = 10^13

C = 47u, so C * dv/dt = 470 * 10^6

V * C * dv/dt = power = 5 * 470 * 10^6 = 2.35 * 10^9 = 2.35GW which matches the graph and is explained by the equations.

Inductor: V = L*di/dt
At peak, v=5.
5/50mH = 100 = di/dt
i starts at 0, and v is held to 5V for 5us. So I/(5us) = 100 and I = 5 * 10^-4
v=5 * i=5*10^-4 == power = 2.5mW
Which matches the graph

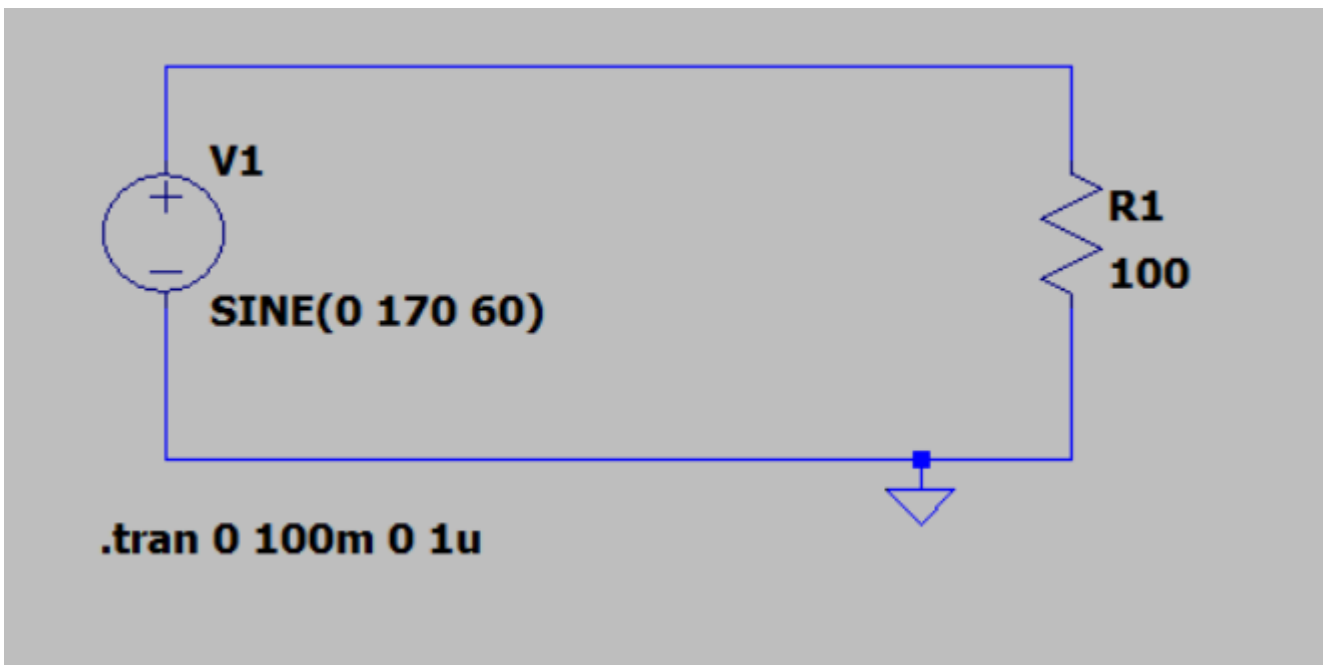
Section 2:

For each of the circuits in Figure 1 we will be studying the response of these passive loads to sinusoidal inputs.

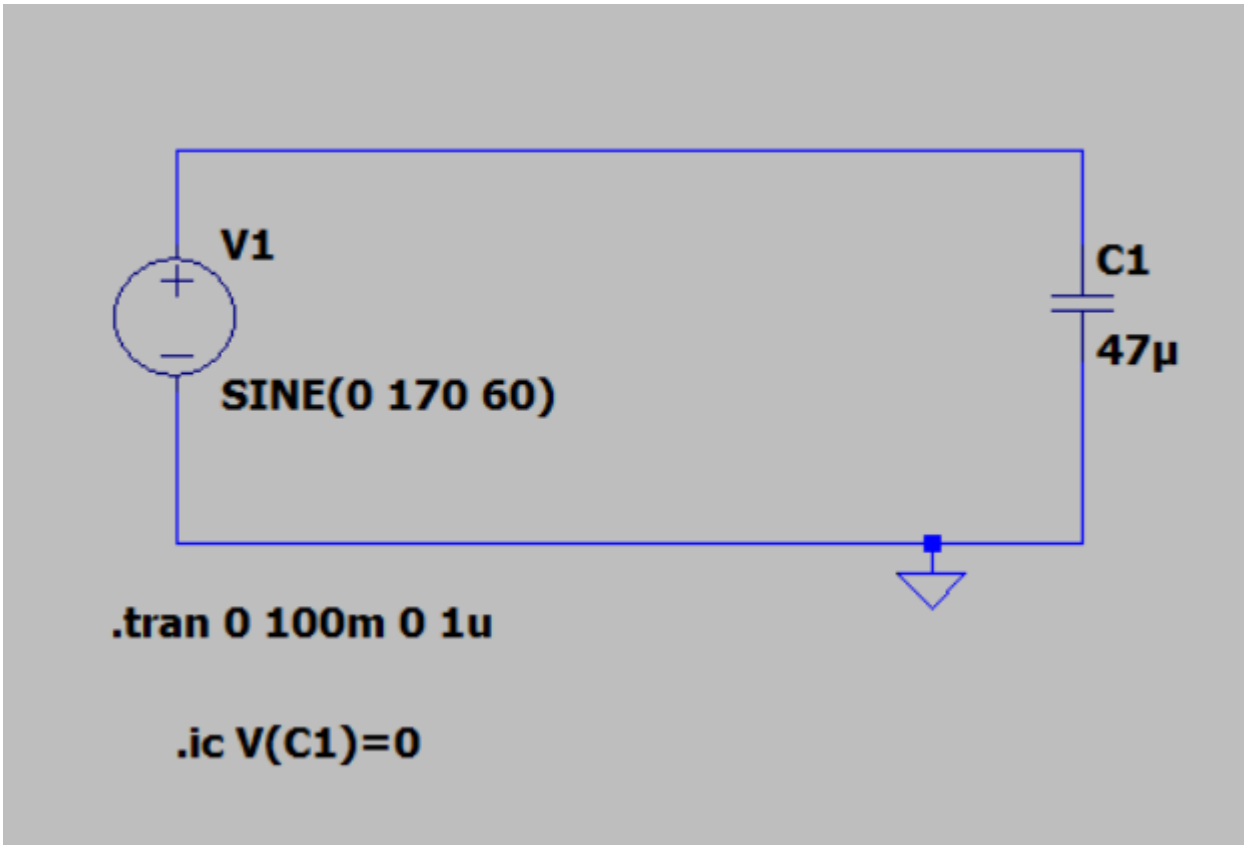
Simulations:

(1) Make the voltage source Vin a sinusoidal source with 170V amplitude and 60Hz frequency.
(2) Run a Transient Analysis for 100ms, using 1us time step.

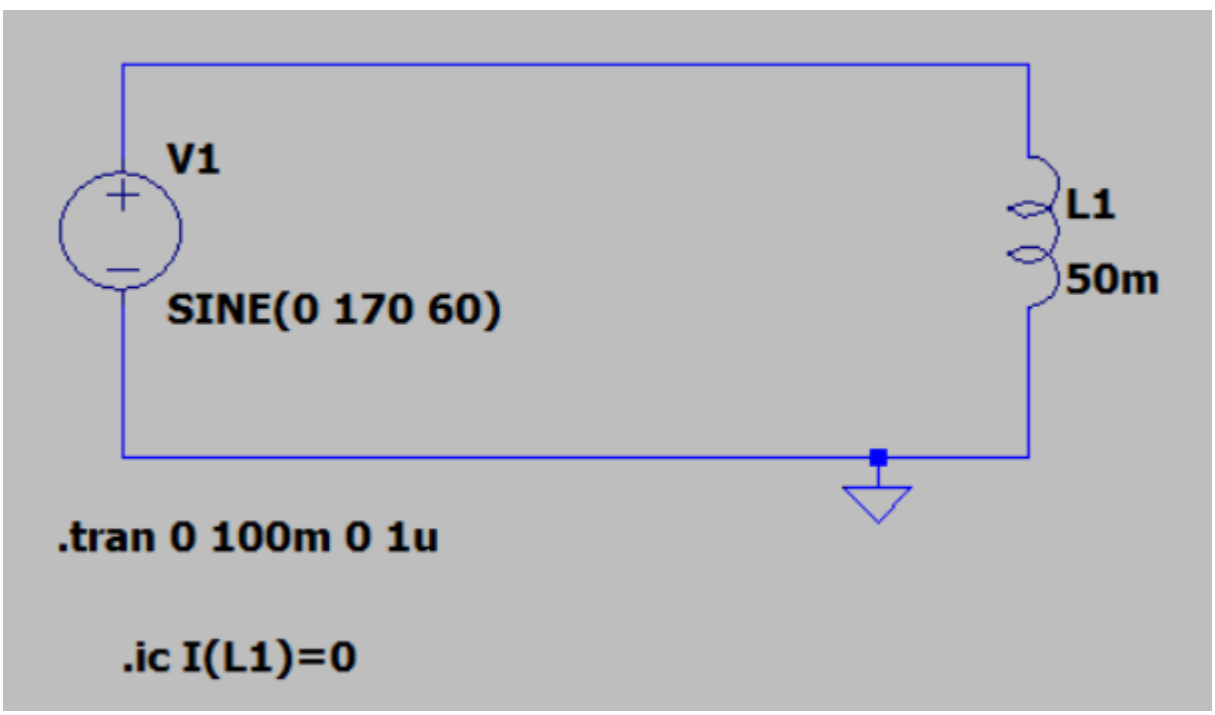
Resistor Circuit:



Capacitor Circuit:

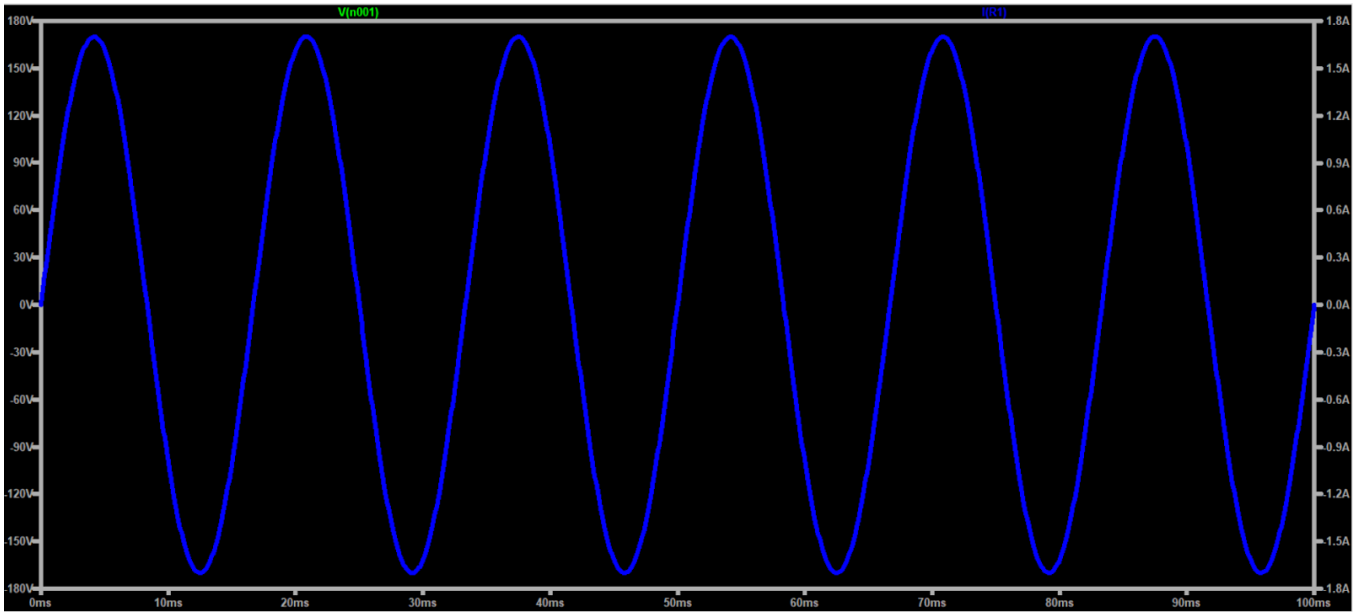


Inductor Circuit:

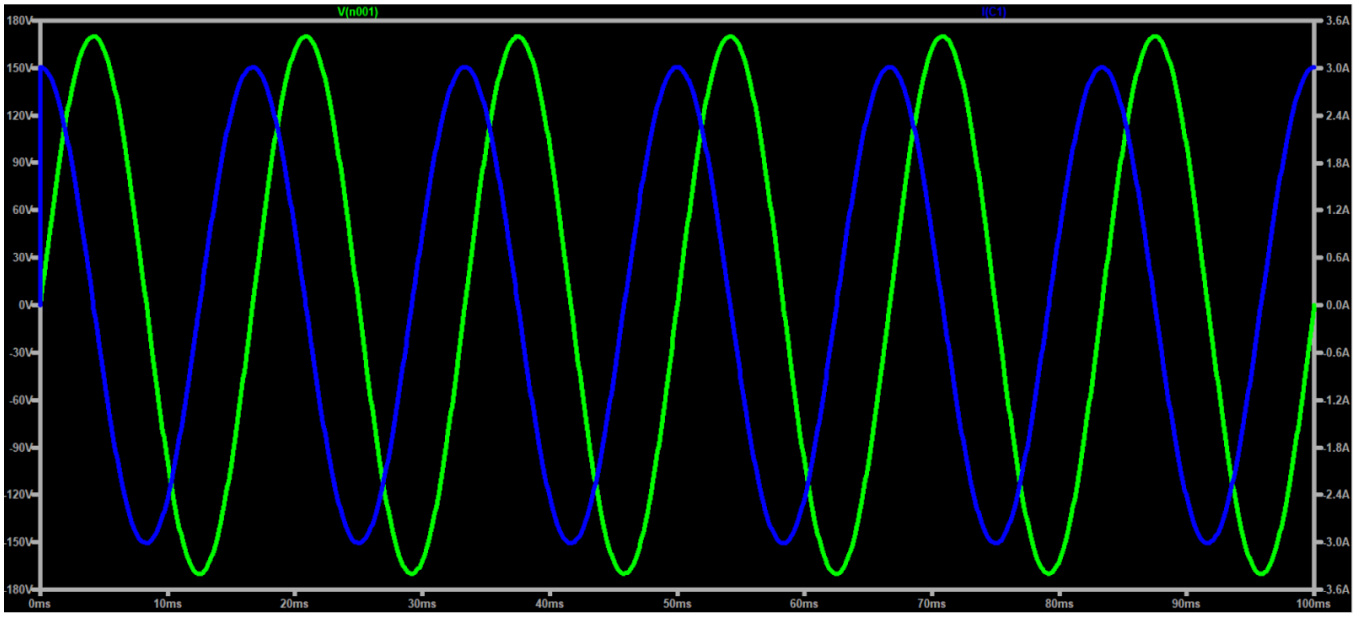


(3) Zoom in on two or three cycles of the output *after* it has reached steady-state.
(4) Plot voltage and current in the passive element.

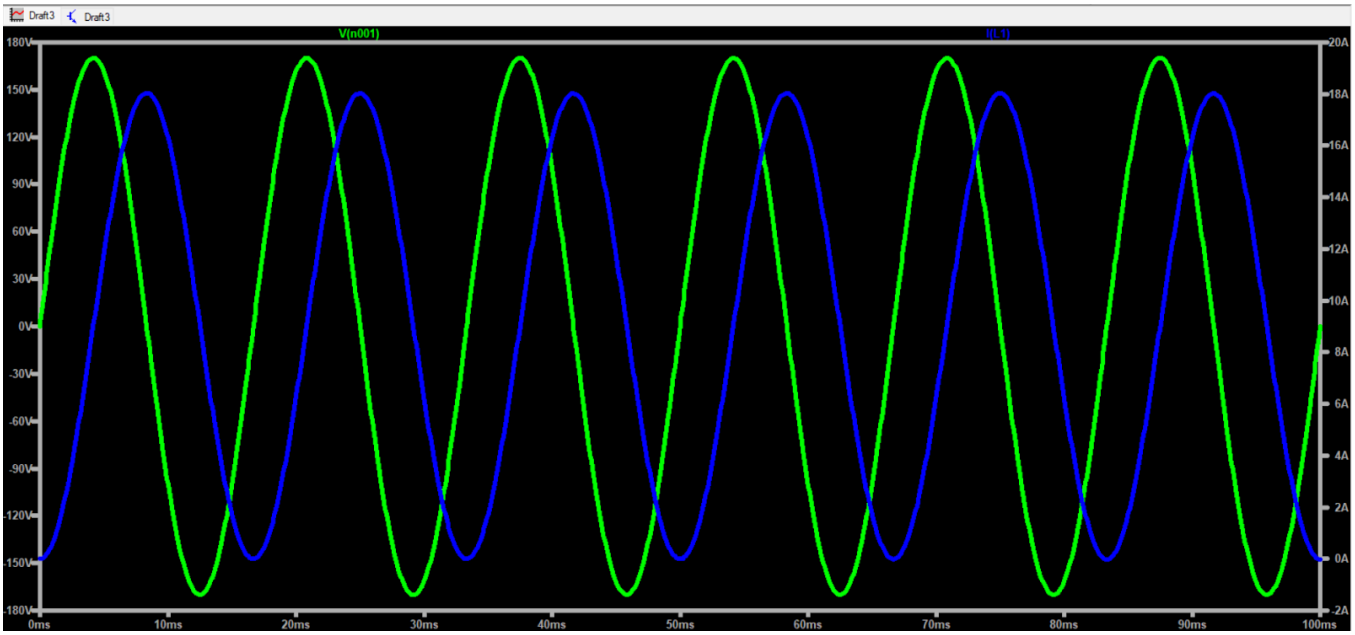
Resistor (current and voltage overlap perfectly)



Capacitor

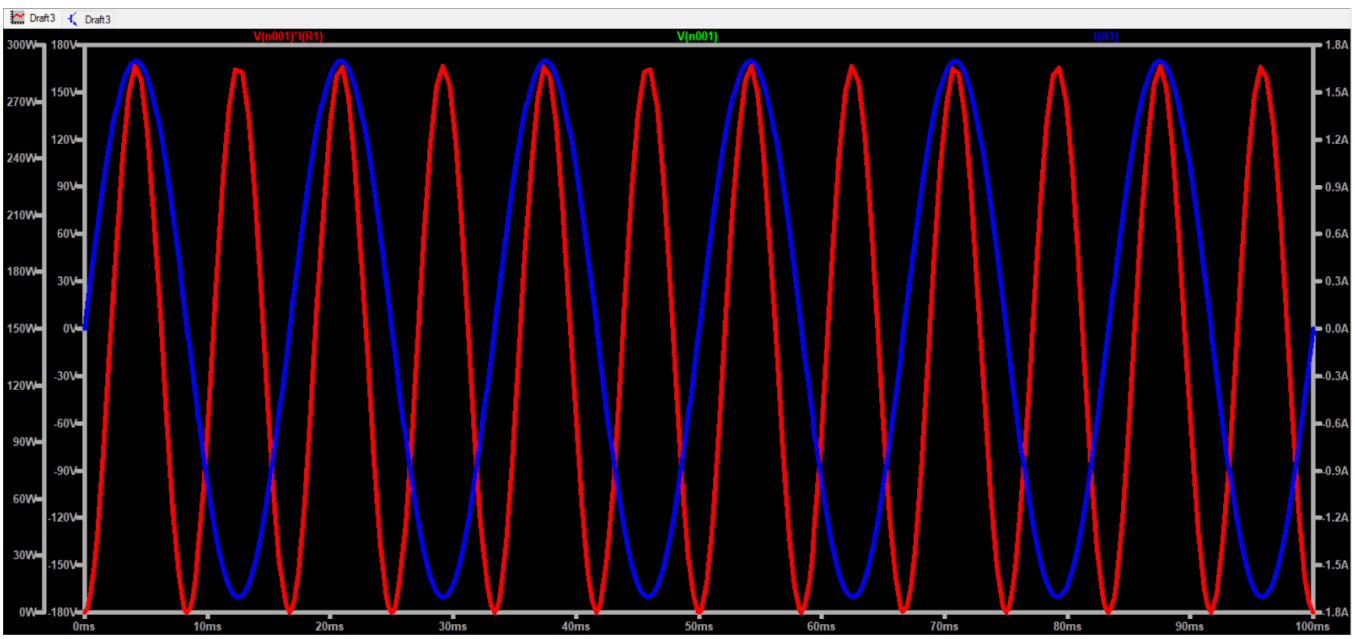


Inductor

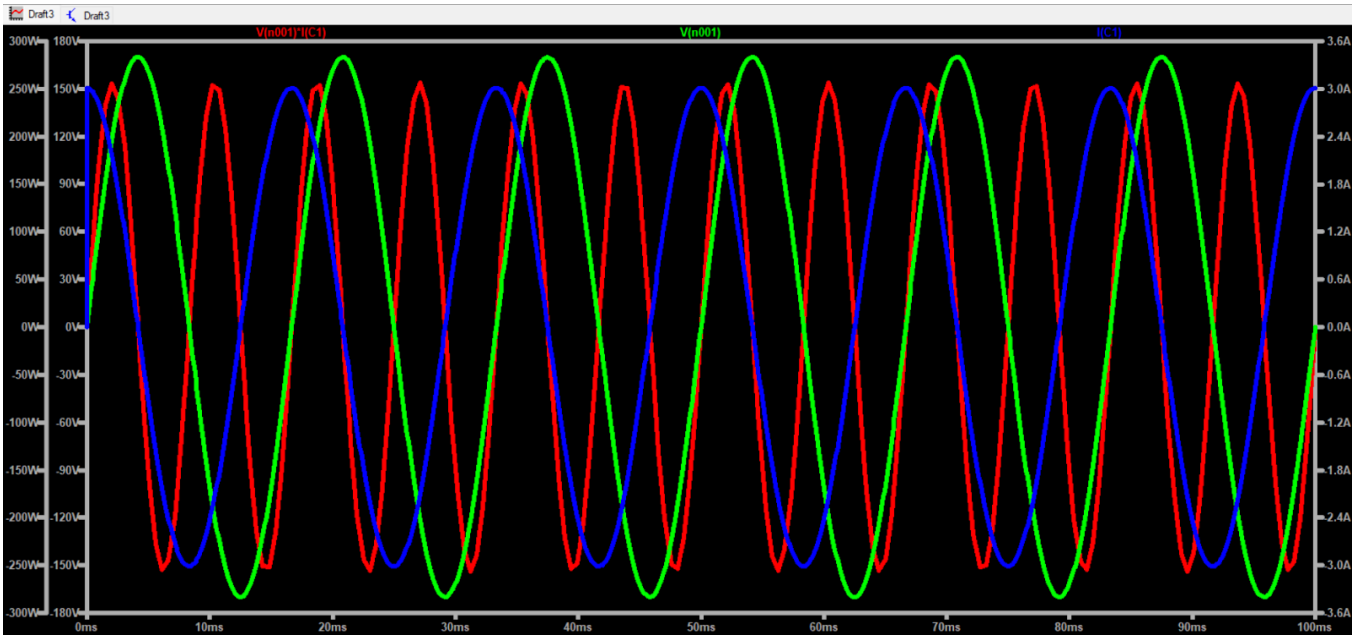


(5) Multiply the voltage and current and observe the instantaneous power.

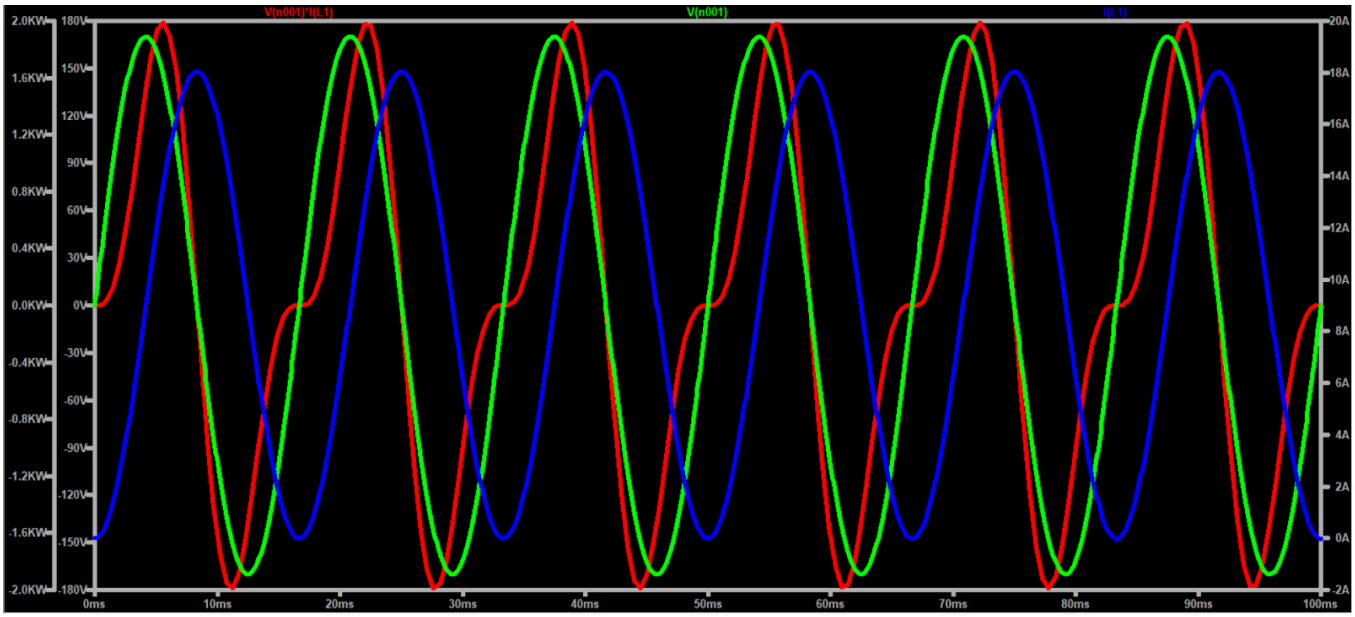
Resistor



Capacitor



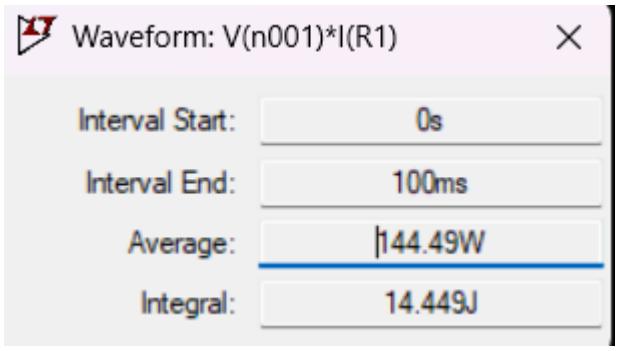
Inductor



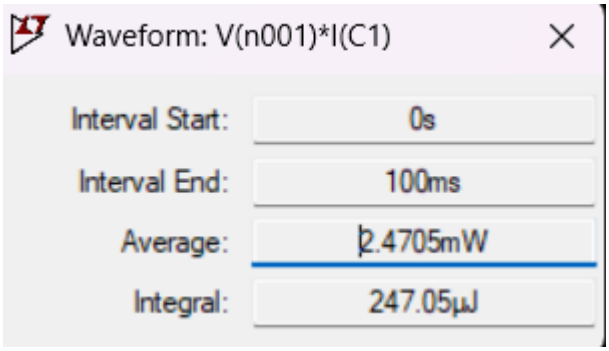
(6) From the instantaneous power in the element, measure the average.

(7) To view average and RMS voltage/current, hold “ctrl” key and click on the name of the waveform of interest.

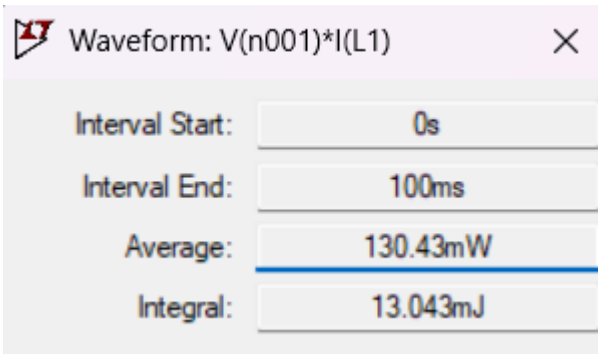
Resistor:



Capacitor



Inductor



Observations:

(1) Observe the current waveform and explain its relationship to the voltage.

The currents follow what we were taught in class with the mnemonic ELI the ICE man. The current and voltage for resistor are perfectly in sync, but for the capacitor, the voltage is shifted behind the current, and for the inductor, the current is shifted behind the voltage.

(2) Define the relationship for current vs. voltage for the element and support your explanations.

Same as section 1.

(3) Explain what you observe in the average power for each load.

The average power for the resistor is an expected large wattage (144) while the capacitor and inductor are close to 0 (130mW and 2.47mW respectively). These numbers are all close to what we expect.

Expected Resistor: 170/sqrt(2)=120.2

120.2/100 = 1.202
120.2 * 1.202 = 144.5W

Expected Capacitor and Inductor =0W

(4) How are they different for the various loads?
Because of PF behavior, only the average power for the resistor is non-zero.

Part 2: Measuring RLCs

Section 1: Known Components

Silver resistor looking thing: 24.651 ohm + 30.95 degrees – inductor
Big thing: 68.678kohm + 80.73 degrees – inductor
Red thing: 168ohm - 88.89 degrees – capacitor
Blue thing: 685ohm - 89.77 degrees – capacitor
Yellow thing: 16.879kohm - 89.65 degrees – capacitor

Section 2: "Mystery Box"
Make measurements on the components in the mystery box (ask your TA). **Don't open the box.** Identify them by doing an impedance and phase measurement (Z and Θ). Then put the appropriate measurement parameters and find the values for the components and their parasitics. Take note of the number marked in the mystery component and to don't forget to add to your report. Once you have all your measurement, open the box and see how the mystery component looks like. Add this picture to your report alongside the number on the box and the values you found.

Number 3: 6.65 ohm + 71.73 degrees – inductor

Report Questions:

- Look back to Part 1 Section 1, where the square wave voltage is applied to the inductor. What do you think would happen to the inductor current waveform if the magnitude of the driving square wave was increased to +/- 10V? What would happen if the inductor value was increased to 10 times its given value?
  If the magnitude was increased, the abs value of the slope of the current would also increase proportionally to the voltage (2x). If the inductor value was increased however, the slope of the current would decrease proportionally to the inductor change (10x decrease).
- Observe the input signal in Part 1 Section 2. What does this represent?
  The voltage wave that wall outlets output.
- From Part 2 what observations can you make about the passives we use in circuits, and what do we need to specifically consider in power circuits?
  To transfer power, we like capacitors and inductors, but if we want to deliver power to a load, then resistors are the best choice.